WEEK 11

Illustration of Array functions

- Count function
- Looping
- Sort





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ARRAY FUNCTION



The array functions allow you to access and manipulate arrays. Simple and multi-dimensional arrays are supported.

The array functions are part of the PHP core. There is no installation needed to use these functions.

```
<?php
$courses=array(“BSIT",“BSHRM",“ACT");
echo count($courses);
?>
```

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ARRAY FUNCTION

Function	Description
array()	Creates an array
array_change_key_case()	Changes all keys in an array to lowercase or uppercase
array_chunk()	Splits an array into chunks of arrays
array_column()	Returns the values from a single column in the input array
array_combine()	Creates an array by using the elements from one "keys" array and one "values" array
array_count_values()	Counts all the values of an array
array_diff()	Compare arrays, and returns the differences (compare values only)
array_diff_assoc()	Compare arrays, and returns the differences (compare keys and values)
array_diff_key()	Compare arrays, and returns the differences (compare keys only)
array_diff_uassoc()	Compare arrays, and returns the differences (compare keys and values, using a user-defined key comparison function)
array_diff_ukey()	Compare arrays, and returns the differences (compare keys only, using a user-defined key comparison function)



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array_fill()	Fills an array with values
array_fill_keys()	Fills an array with values, specifying keys
array_filter()	Filters the values of an array using a callback function
array_flip()	Flips/Exchanges all keys with their associated values in an array
array_intersect()	Compare arrays, and returns the matches (compare values only)
array_intersect_assoc()	Compare arrays and returns the matches (compare keys and values)
array_intersect_key()	Compare arrays, and returns the matches (compare keys only)
array_intersect_uassoc()	Compare arrays, and returns the matches (compare keys and values, using a user-defined key comparison function)
array_intersect_ukey()	Compare arrays, and returns the matches (compare keys only, using a user-defined key comparison function)
array_key_exists()	Checks if the specified key exists in the array
array_keys()	Returns all the keys of an array
array_map()	Sends each value of an array to a user-made function, which returns new values
array_merge()	Merges one or more arrays into one array
array_merge_recursive()	Merges one or more arrays into one array recursively
array_multisort()	Sorts multiple or multi-dimensional arrays
array_pad()	Inserts a specified number of items, with a specified value, to an array
array_pop()	Deletes the last element of an array

array_product()	Calculates the product of the values in an array
array_push()	Inserts one or more elements to the end of an array
array_rand()	Returns one or more random keys from an array
array_reduce()	Returns an array as a string, using a user-defined function
array_replace()	Replaces the values of the first array with the values from following arrays
array_replace_recursive()	Replaces the values of the first array with the values from following arrays recursively
array_reverse()	Returns an array in the reverse order
array_search()	Searches an array for a given value and returns the key
array_shift()	Removes the first element from an array, and returns the value of the removed element
array_slice()	Returns selected parts of an array
array_splice()	Removes and replaces specified elements of an array
array_sum()	Returns the sum of the values in an array
array_udiff()	Compare arrays, and returns the differences (compare values only, using a user-defined key comparison function)
array_udiff_assoc()	Compare arrays, and returns the differences (compare keys and values, using a built-in function to compare the keys and a user-defined function to compare the values)

array_udiff_uassoc()	Compare arrays, and returns the differences (compare keys and values, using two user-defined key comparison functions)
array_uintersect()	Compare arrays, and returns the matches (compare values only, using a user-defined key comparison function)
array_uintersect_assoc()	Compare arrays, and returns the matches (compare keys and values, using a built-in function to compare the keys and a user-defined function to compare the values)
array_uintersect_uassoc()	Compare arrays, and returns the matches (compare keys and values, using two user-defined key comparison functions)
array_unique()	Removes duplicate values from an array
array_unshift()	Adds one or more elements to the beginning of an array
array_values()	Returns all the values of an array
array_walk()	Applies a user function to every member of an array
array_walk_recursive()	Applies a user function recursively to every member of an array
arsort()	Sorts an associative array in descending order, according to the value
asort()	Sorts an associative array in ascending order, according to the value
compact()	Create array containing variables and their values

count()	Returns the number of elements in an array
current()	Returns the current element in an array
each()	Deprecated from PHP 7.2. Returns the current key and value pair from an array
end()	Sets the internal pointer of an array to its last element
extract()	Imports variables into the current symbol table from an array
in_array()	Checks if a specified value exists in an array
key()	Fetches a key from an array
krsort()	Sorts an associative array in descending order, according to the key
ksort()	Sorts an associative array in ascending order, according to the key
list()	Assigns variables as if they were an array
natcasesort()	Sorts an array using a case insensitive "natural order" algorithm
natsort()	Sorts an array using a "natural order" algorithm
next()	Advance the internal array pointer of an array
pos()	Alias of current()
prev()	Rewinds the internal array pointer
range()	Creates an array containing a range of elements
reset()	Sets the internal pointer of an array to its first element
rsort()	Sorts an indexed array in descending order
shuffle()	Shuffles an array
sizeof()	Alias of count()
sort()	Sorts an indexed array in ascending order
uasort()	Sorts an array by values using a user-defined comparison function
uksort()	Sorts an array by keys using a user-defined comparison function
usort()	Sorts an array using a user-defined comparison function

ARRAY FUNCTION



Array change key case

Changes all keys in an array to lowercase or uppercase

```
<?php
$age=array("student1"=>"90","student2"=>"87","student3"=>"93");
print_r(array_change_key_case($age,CASE_UPPER));
?>
```

Parameter	Description
array	Required. Specifies the array to use
case	•Optional. Possible values:CASE_LOWER - Default value. Changes the keys to lowercase •CASE_UPPER - Changes the keys to uppercase

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ARRAY FUNCTION



Array chunk function

The array_chunk() function splits an array into chunks of new arrays

```
<?php
$cars=array("Volvo","BMW","Toyota","Honda","Mercedes","Opel");
print_r(array_chunk($cars,2));
?>
```

Parameter	Description
array	Required. Specifies the array to use
size	Required. An integer that specifies the size of each chunk
preserve_key	•Optional. Possible values:true - Preserves the keys •false - Default. Reindexes the chunk numerically

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ARRAY FUNCTION

Array column

The array_column() function returns the values from a single column in the input array.

```
<?php
$a = array(
  array(
    'id' => 5698,
    'first_name' => 'Peter',
    'last_name' => 'Griffin', ),
  array(
    'id' => 4767,
    'first_name' => 'Ben',
    'last_name' => 'Smith', )
);
$last_names = array_column($a, 'last_name');
print_r($last_names);
?>
```



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Parameter	Description
array	Required. Specifies the multi-dimensional array (record-set) to use. As of PHP 7.0, this can also be an array of objects.
column_key	Required. An integer key or a string key name of the column of values to return. This parameter can also be NULL to return complete arrays (useful together with index_key to re-index the array)
index_key	Optional. The column to use as the index/keys for the returned array

ARRAY FUNCTION



Array combine

The array_combine() function creates an array by using the elements from one "keys" array and one "values" array.

```
<?php
$fname=array("Peter","Ben","Joe");
$age=array("35","37","43");

$c=array_combine($fname,$age);
print_r($c);
?>
```

Parameter	Description
keys	Required. Array of keys
values	Required. Array of values

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ARRAY FUNCTION

Array count

The array_count_values() function counts all the values of an array




```
<?php
$a=array("A","Cat","Dog","A","Dog");
print_r(array_count_values($a));
?>
```

Parameter	Description
array	Required. Specifying the array to count values of

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ARRAY FUNCTION



Array diff

The array_diff() function compares **the values** of two (or more) arrays, and returns the differences.

This function compares the values of two (or more) arrays, and return an array that contains the entries from *array1* that are not present in *array2* or *array3*, etc.

```
<?php
$a1=array("a"=>"red","b"=>"green","c"=>"blue","d"=>"yellow");
$a2=array("e"=>"red","f"=>"green","g"=>"blue");

$result=array_diff($a1,$a2);
print_r($result);
?>
```

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ARRAY FUNCTION

- **Array intersect**

  he array_intersect() function compares **the values** of two (or more) arrays, and returns the matches.

  This function compares the values of two or more arrays, and return an array that contains the entries from *array1* that are present in *array2*, *array3*, etc.




```
<?php
$a1=array("a"=>"red","b"=>"green","c"=>"blue","d"=>"yellow");
$a2=array("e"=>"red","f"=>"green","g"=>"blue");

$result=array_intersect($a1,$a2);
print_r($result);
?>
```

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ARRAY FUNCTION



Array key

The array_key_exists() function checks an array for a specified key, and returns true if the key exists and false if the key does not exist.

```
<?php
$a=array("Volvo"=>"XC90","BMW"=>"X5");
if (array_key_exists("Volvo",$a))
  { echo "Key exists!";}
else
  {echo "Key does not exist!";}
?>
```

Parameter	Description
key	Required. Specifies the key
array	Required. Specifies an array

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ARRAY FUNCTION



Array map

The array_map() function sends each value of an array to a user-made function, and returns an array with new values, given by the user-made function.

```
<?php
function myfunction($v)
{
  return($v*$v);
}
$a=array(1,2,3,4,5);
print_r(array_map("myfunction",$a));
?>
```

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ARRAY FUNCTION



Array merge

The array_merge() function merges one or more arrays into one array.

```
<?php
$a1=array("red","green");
$a2=array("blue","yellow");
print_r(array_merge($a1,$a2));
?>
```

Parameter	Description
array1	Required. Specifies an array
array2	Optional. Specifies an array
array3,...	Optional. Specifies an array

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ARRAY FUNCTION

- **Array multisort**

  The array_multisort() function returns a sorted array. You can assign one or more arrays. The function sorts the first array, and the other arrays follow, then, if two or more values are the same, it sorts the next array, and so on.




```
<?php
$a=array("Dog","Cat","Horse","Bear","Zebra");
array_multisort($a);
print_r($a);
?>
```

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ARRAY

PHP - SORT FUNCTIONS FOR ARRAYS



The elements in an array can be sorted in alphabetical or numerical order, descending or ascending.

In this chapter, we will go through the following PHP array sort functions:

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ARRAY

PHP - SORT FUNCTIONS FOR ARRAYS

sort() - sort arrays in ascending order
rsort() - sort arrays in descending order
asort() - sort associative arrays in ascending order, according to the value
ksort() - sort associative arrays in ascending order, according to the key
arsort() - sort associative arrays in descending order, according to the value
krsort() - sort associative arrays in descending order, according to the key



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SORT ARRAY IN ASCENDING ORDER - SORT()

The following example sorts the elements of the $letters array in ascending alphabetical order:

```
<?php
$letters=array(“C",“W",“A");
sort($letters);
?>
```

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ARRAY



SORT ARRAY IN ASCENDING ORDER - SORT()

The following example sorts the elements of the $grade array in ascending numerical order:

```
<?php
$grade=array(86,78,66,75,98);
sort($grade);
?>
```

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ARRAY

SORT ARRAY IN DESCENDING ORDER - RSORT()



The following example sorts the elements of the $letters array in descending alphabetical order:

```
<?php
$letters=array(“C",“W",“A");
rsort($letters);
?>
```

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ARRAY

SORT ARRAY IN DESCENDING ORDER - RSORT()



The following example sorts the elements of the $grade array in descending numerical order:

```
<?php
$grade=array(86,78,66,75,98);
rsort($grade);
?>
```

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ARRAY



SORT ARRAY IN ASCENDING ORDER ACCORDING TO VALUE – asort()

The following example sorts an associative array in ascending order, according to the value:

```
<?php
$score=array(“Math"=>“55",“English"=>“100",“FIL"=>“
65");
asort($score);
?>
```

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ARRAY



SORT ARRAY IN ASCENDING ORDER ACCORDING TO KEY – ksort()

The following example sorts an associative array in ascending order, according to the key:

```
<?php
$score=array(“Math"=>“55",“English"=>“100",“FIL"=>“
65");
ksort($score);
?>
```

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ARRAY



SORT ARRAY IN DESCENDING ORDER ACCORDING TO VALUE – asort()

The following example sorts an associative array in descending order, according to the value:

```
<?php
$score=array(“Math"=>“55",“English"=>“100",“FIL"=>“
65");
arsort($score);
?>
```

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ARRAY



SORT ARRAY IN DESCENDING ORDER ACCORDING TO KEY – ksort()

The following example sorts an associative array in descending order, according to the key:

```
<?php
$score=array(“Math"=>“55",“English"=>“100",“FIL"=>“
65");
krsort($score);
?>
```

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asort() - sort associative arrays in ascending order, according to the value
ksort() - sort associative arrays in ascending order, according to the key
arsort() - sort associative arrays in descending order, according to the value
krsort() - sort associative arrays in descending order, according to the key



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